RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Even Semester)

Innovative Teaching Method Execution

UNIT I Amplitude Modulation

Degree, Semester& Branch: IV Semester B.E.ECE A
Course Code & Title: EC8491 Communication Theory
Name of the Faculty member: Mrs.G.Gnana Priya
Name of the Topic: Need for modulation
Name of the Innovative Practice: Role Play
Date & Duration: 19.12.2019 & 15 Minutes

Description of Role Play Activity:

Role play is a technique that allows students to explore realistic situations by interacting with other people in a managed way in order to develop experience and trial different strategies in a supported environment. It is widely agreed that learning takes place when activities are engaging and memorable.

Goals (Learning Outcomes):

- ❖ The students will be able to acquire knowledge about the need for modulation.

Use of appropriate methods:

Justification for choosing the topic using role play activity:

This activity helps to develop the way of thinking of the students and they will be able to visualize the nature of the modulating signal, carrier signal and the process of modulation. As an electronics and communication engineering student modulation is an important concept and they should know about it clearly. So I had chosen this topic for conducting role play activity.

Effective presentation

Details of the Implementation:

- ➢ First I asked the students to discuss among them about the concept of modulation
- ➢ Then they were asked to come to the stage as group
- ➢ Each group consisted of 4 students
- ➢ One student acted as the transmitter, one as receiver, one as message signal and the other one as carrier signal
- ➢ After assigning the role they presented the concept of modulation visually

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

After conducting this activity, the students got a clear idea about the modulation process and the need for modulation. The effectiveness of the activity was felt while asking questions in the classroom and expecting the same in the Internal Assessment Test also.

Reflective Critique:

- **Benefits of conducting this activity:**

  Since this activity was conducted for groups, it improves their collaborative learning. The concept of modulation reached all the students as it is presented visually.

- **Challenges:**

  Initially the students got hesitated to come to the dias and present. Then I encouraged them to come forward. So it took around 5 minutes to make the students to come forward and present.

Samples images of the activity:

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1	2			2		2		2

CO1: The students will be able design AM communication systems.

References:

1. J.G.Proakis, M.Salehi, "Fundamentals of Communication Systems", Pearson Education 2014
2. Simon Haykin, -"Communication Systems", 4th Edition, Wiley, 2014
3. https://ocw.mit.edu/resources/ res-6 -007-signals- and- systems- spring -2011/ video-lectures/ lecture-14-demonstration-of-amplitude-modulation

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<u>Innovative Teaching Method Execution</u>

<u>UNIT I Amplitude Modulation</u>

Degree, Semester& Branch: IV Semester B.E.ECE A
Course Code & Title: EC8491 Communication Theory
Name of the Faculty member: Mrs.G.Gnana Priya
Name of the Topic: Comparison of different AM techniques
Name of the Innovative Practice: Mind Map
Date & Duration: 08.01.2020 & 15 Minutes

Description of Mind Map Activity:

A mind map is a graphical way to represent ideas and concepts. It is a visual thinking tool that helps structuring information, helping the students to better analyze, comprehend, synthesize, recall and generate new ideas. It is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

<u>Goals (Learning Outcomes):</u>

- The students will be able to compare the various amplitude modulation techniques.

<u>Use of appropriate methods:</u>

Justification for choosing the topic using mind map activity:

Since comparison of different amplitude modulation techniques is an important topic in concept wise and university exam point of view, I have chosen this topic and the reason for choosing mind map activity is while recollecting the concepts and hints learnt in the classroom, the students will be able to reproduce the concept in the exam.

<u>Effective presentation</u>

Details of the Implementation:

The students were randomly called and they were asked to write any one comparison of different AM techniques like bandwidth required, power efficiency, receiver complexity etc., that they had learnt in the classroom as a pictorial representation in the white board.

<u>Significance of Results:</u>

Assessment of Effectiveness/Success of the Activity:

Due to this activity, the students were able to recall the topics they had learnt in the classroom and they were also able to reproduce it in the exam.

Reflective Critique:

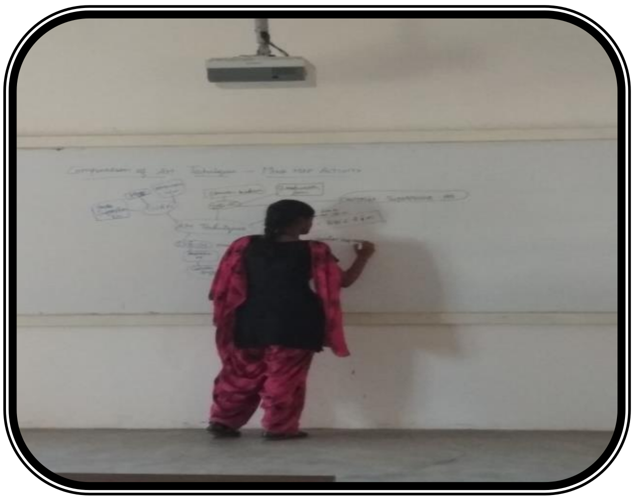
- **Benefits of conducting this activity:**

  The students gave oral feedback that now they got a clear idea about the classification of various amplitude modulation techniques, and they asked me to conduct more such activities like this.

- **Challenges:**

  I planned to conduct this activity within 10 minutes. But it took 15 minutes to finish this activity, since few students got hesitate to come to the board and write, I made those students to come forward. So it took 5 minutes extra time to finish this activity.

Samples images of the activity:



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1	2			2		2		2

CO1: The students will be able design AM communication systems.

References:

1. J.G.Proakis, M.Salehi, "Fundamentals of Communication Systems", Pearson Education 2014
2. Simon Haykin, -"Communication Systems", 4th Edition, Wiley, 2014

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Academic Year: 2019- 2020 (Even Semester)

Innovative Practices Description

UNIT II Angle Modulation

Degree, Semester& Branch: IV Semester B.E.ECE A
Course Code & Title: EC8491 Communication Theory
Name of the Faculty member: Mrs.G.Gnana Priya
Name of the Topic: Narrowband and Wideband FM
Name of the Innovative Practice: Zero Minute Speech
Date & Duration: 20.01.20 & 15 minutes

Description of Zero Minute Speech:

Zero minute speech activity makes the student to come forward and share the things they have learnt during that particular class. Due to this activity the student will be able to recall the topic learnt in the classroom and they can reproduce it in the exam. It also enables the students to explain about the topic in short which can be understood by other students easily.

Goals(Learning Outcome):

- The students will be able to compare narrowband frequency modulation with wideband frequency modulation.
- The students will also be able to elaborate in detail about the angle modulation techniques

Use of appropriate methods:

Justification for choosing the topic using Zero Minute Speech Activity:

Since this activity is used to analyse the understanding level of the students within the classroom in a better way by asking the students to come forward and explain, I have chosen this topic to conduct this activity. In university exam point of view, the comparison of narrowband frequency modulation with wideband frequency modulation is important. So that is also another reason why I have chosen this topic to conduct zero minute speech activity.

Effective Presentation:

Details of Implementation:

- ❖ At the beginning of the class I informed the students that I am going to conduct zero minute speech activity at the end of that particular class
- ❖ The students were asked to recall the topic learnt in the classroom and they were asked to give a speech for few minutes about the narrowband and wideband FM
- ❖ Around 7 students were randomly called and they were asked to talk about any two comparisons.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ Due to this activity, the active learning skills of the students got increased.
- ❖ The assessment of effectiveness of the activity was felt when I asked questions in the classroom and expecting the same in Internal Assessment Tests also.

Sample Images:

Reflective Critique:

- **Benefits of conducting this activity**:
  - ✓ Since the conduct of this activity was informed to the students at the beginning of the particular class, it enables them to listen to the class carefully
  - ✓ Without any prior preparation, the students can easily explain about the particular topic by listening to the concept in the classroom.
  - ✓ By conducting this activity it was easy for the students to recollect the points learnt in the classroom. They will also be able to reproduce the concepts in the exam.

- **Challenges faced in conducting this activity**:

  I planned to conduct this activity within 5 minutes. But it took around 15 minutes to conduct this activity, since few students hesitate to come to the dias and share the concept. I changed their mentality and brought them to the stage to share the concept. Only very few students got chance to come and share their ideas since the time duration for conducting this activity is limited.

Changes required for future:

Since the students feel shy to come to the dias and share the concept, next time while conducting this activity I have planned to encourage the students by giving gifts like pen, chocolates etc.,

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	2	1	1	2			2		2		2

CO2: The students will be able to elaborate the angle modulation and demodulation techniques

References:

1. K.N.Hari Bhat, D.Ganesh Rao - Principles of Communication Systems, Cengage, 2017
2. J.G.Proakis, M.Salehi, “Fundamentals of Communication Systems”, Pearson Education 2014.